CSE 102 (Fall 2021) – Homework 2

Due: Oct 12 at 10pm PT on CrowdGrader.

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1. Let $f(n)$ be a positive, increasing function that satisfies $f(\frac{n}{2}) = \Theta(f(n))$. Prove that $\Sigma_{i=1}^{n} f(i) = \Theta(n \cdot f(n))$.

$$\Sigma_{i=1}^{n} f(i) = \Theta(n \cdot f(n)) \tag{1}$$

Solution.

$$\Sigma_{i=1}^{n} f(i) \leq \Sigma_{i=1}^{n} f(n) \Rightarrow n \cdot f(n) \Rightarrow O(n \cdot f(n)) \tag{2}$$

$$\Sigma_{i=1}^{n} f(i) \geq \Sigma_{i=\lceil\frac{n}{2}\rceil}^{n} f(i) \tag{3}$$

$$\Rightarrow \Sigma_{i=\lceil\frac{n}{2}\rceil}^{n} f\left(\lceil\frac{n}{2}\rceil\right) \Rightarrow \left(n - \lceil\frac{n}{2}\rceil\right) \cdot f\left(\lceil\frac{n}{2}\rceil\right)$$

$$\Rightarrow \left(n - \lceil\frac{n}{2}\rceil + 1\right) \cdot f\left(\lceil\frac{n}{2}\rceil\right) > \left(\frac{n}{2}\right) \cdot f\left(\frac{n}{2}\right)$$

$$\Rightarrow \left(\frac{n}{2}\right) \cdot f\left(\frac{n}{2}\right) \approx n \cdot f(n) \Rightarrow \Omega(n \cdot f(n))$$

$$\textit{Because: } f(\frac{n}{2}) = \Theta(f(n)) \textit{ Therefore:}$$

$$O(n \cdot f(n)) \cap \Omega(n \cdot f(n)) = \Theta(n \cdot f(n)) \tag{4}$$

2. Use the result of the preceding problem to prove that $log(n!) = \Theta(n \cdot log(n))$, without using Stirling's formula.

$$log(n!) = \Theta(n \cdot log(n)) \tag{5}$$

Solution.

$$log(n!) = log(1) + log(2) + \cdots + log(n) = \Sigma_{i=1}^{n} log(i) \tag{6}$$

Using the result of Q1 where $\Sigma_{i=1}^{n} f(i) = \Theta(n \cdot f(n))$ *then:*

$$log(n!) = \Sigma_{i=1}^{n} log(i) = \Theta(n \cdot log(n)) \tag{7}$$

$$\tag{8}$$

3. Use Stirling's formula to determine a constant $a > 0$ such that $\binom{3n}{n} = \Theta(\frac{a^n}{\sqrt{n}})$.

$$\binom{3n}{n} = \Theta(\frac{a^n}{\sqrt{n}}) \tag{9}$$

Stirling's formula is given by:

$$n \in Z^+ \tag{10}$$
$$n! = \sqrt{2\pi n} \cdot \left(\frac{n}{e}\right)^n \cdot (1 + \Theta(\frac{1}{n})) \tag{11}$$
$$\tag{12}$$

Solution.

$$\binom{3n}{n} = \frac{3n!}{n!2n!} \tag{13}$$

$$\Rightarrow \frac{\sqrt{2\pi 3n} \cdot \left(\frac{3n}{e}\right)^{3n} \cdot (1 + \Theta(\frac{1}{3n}))}{[\sqrt{2\pi n} \cdot \left(\frac{n}{e}\right)^n \cdot (1 + \Theta(\frac{1}{n}))] \cdot [\sqrt{2\pi 2n} \cdot \left(\frac{2n}{e}\right)^{2n} \cdot (1 + \Theta(\frac{1}{2n}))]} \tag{14}$$

$$\Rightarrow \frac{\sqrt{2\pi} \cdot \sqrt{3n} \cdot \frac{3n^{3n}}{e^{3n}} \cdot (1 + \Theta(\frac{1}{3n}))}{[\sqrt{2\pi} \cdot \sqrt{n} \cdot \frac{n^n}{e^n} \cdot (1 + \Theta(\frac{1}{n}))] \cdot [\sqrt{2\pi} \cdot \sqrt{2n} \cdot \frac{2n^{2n}}{e^{2n}} \cdot (1 + \Theta(\frac{1}{2n}))]} \tag{15}$$

$$\Rightarrow \frac{\sqrt{2\pi}}{\sqrt{2\pi}^2} \cdot \frac{\sqrt{3n}}{\sqrt{n} \cdot \sqrt{2n}} \cdot \frac{\frac{3^{3n} \cdot n^{3n}}{e^{3n}}}{\frac{n^n}{e^n} \cdot \frac{2^{2n} \cdot n^{2n}}{e^{2n}}} \cdot \left[\frac{1 + \Theta(\frac{1}{3n})}{1 + \Theta(\frac{1}{n}) \cdot 1 + \Theta(\frac{1}{2n})}\right] \tag{16}$$

$$\Rightarrow \frac{1}{\sqrt{2\pi}} \cdot \frac{\sqrt{3}}{\sqrt{2n}} \cdot \frac{3^{3n}}{2^{2n}} \cdot \left[\frac{1 + \Theta(\frac{1}{3n})}{1 + \Theta(\frac{1}{n}) \cdot 1 + \Theta(\frac{1}{2n})}\right] \tag{17}$$

$$\Rightarrow \frac{\sqrt{3}}{\sqrt{4\pi}} \cdot \frac{\frac{9}{4}^n}{\sqrt{n}} \cdot \left[\frac{1 + \Theta(\frac{1}{3n})}{1 + \Theta(\frac{1}{n}) \cdot 1 + \Theta(\frac{1}{2n})}\right] \tag{18}$$

$$\Rightarrow \frac{\binom{3n}{n}}{\frac{\frac{9}{4}^n}{\sqrt{n}}} = \frac{\sqrt{3}}{\sqrt{4\pi}} \cdot \left[\frac{1 + \Theta(\frac{1}{3n})}{1 + \Theta(\frac{1}{n}) \cdot 1 + \Theta(\frac{1}{2n})}\right] \tag{19}$$

Since $0 < \frac{\sqrt{3}}{\sqrt{4\pi}} < \infty$ *Then:*

$$\binom{3n}{n} = \Theta\left(\frac{\frac{9}{4}^n}{\sqrt{n}}\right) \tag{20}$$

The coefficient a is given above as $a = \frac{9}{4}$.

4. Define $S(n)$ for $n \in Z^+$ by the recurrance

$$S(n) = \begin{cases} 0, & \text{if } n = 1. \\ S(\lceil \frac{n}{2} \rceil) + 1, & \text{if } n \geq 2. \end{cases} \tag{21}$$

Prove that $S(n) \geq log(n)$ for $n \geq 1$. Hence $S(n) = \Omega(log(n))$.

Solution.

$$S(n) = S(\lceil \frac{n}{2} \rceil) + 1 \tag{22}$$

$$= (S(\frac{n}{2^2}) + 1) + 1 = S(\frac{n}{2^2}) + 2$$

$$= (S(\frac{n}{2^3}) + 2) + 1 = S(\frac{n}{2^3}) + 3$$

$$= S(\frac{n}{2^k}) + k$$

$$S(n) = S(\frac{n}{2^k}) + k \tag{23}$$

$$\textit{If we assume } \frac{n}{2^k} = 1 \textit{ Then:}$$

$$\frac{n}{2^k} = 1 \Rightarrow n = 2^k \tag{24}$$

$$\Rightarrow k = \log_2(n)$$

$$S(n) = S(1) + \log_2(n) = 0 + \log_2(n) \tag{25}$$

$$\Rightarrow S(n) = \Theta(\log_2(n)) \tag{26}$$

$$\textit{Because } \Omega(g(n)) \in \Theta(g(n)) \Rightarrow S(n) = \Omega(\log_2(n)) \tag{27}$$

5. Let $T(n)$ be defined by the recurrence

$$T(n) = \begin{cases} 1, & \text{if } n = 1. \\ T(\lfloor \frac{n}{2} \rfloor) + n^2, & \text{if } n \geq 2. \end{cases} \tag{28}$$

Show that for all $n \geq 1$, $T(n) \leq \frac{4}{3} \cdot n^2$. Hence $T(n) = O(n^2)$.

Solution.

$$\begin{aligned} T(n) &= T(\frac{n}{2}) + n^2 \qquad (29) \\ &= T(\frac{n}{2^2}) + \left(\frac{n}{2}\right)^2 + n^2 \\ &= T(\frac{n}{2^3}) + \left(\frac{n}{2^2}\right)^2 \left(\frac{n}{2}\right)^2 + n^2 \\ T(n) &= T(\frac{n}{2^k}) + \frac{n}{2^{k-1}}^2 \qquad (30) \\ k &= \log_2(n) - 1 \qquad (31) \\ \Rightarrow &= n^2(1 + \frac{1}{4} + \frac{1}{4}^2 + \cdots + \frac{1}{4}^{\log_2(n)-1}) \qquad (32) \\ \Rightarrow &= n^2 \cdot \Sigma_{i=0}^{\log_2(n)-1} \frac{1}{4}^i \to n^2 \cdot \frac{1}{1-\frac{1}{4}} = \frac{1}{\frac{3}{4}} \cdot n^2 = \frac{4}{3} \cdot n^2 \qquad (33) \\ T(n) &= \frac{4}{3} \cdot n^2 \Rightarrow T(n) = \Theta(n^2) \Rightarrow O(n^2) \qquad (34) \\ & \qquad (35) \end{aligned}$$

By using the fact that the recurrence generates a Geometric series, we can use the sum equation for the Geometric Series to reach the above result that $T(n) = \Theta(n^2)$ since $T(n) = \frac{4}{3} \cdot n^2$.

6. Prove that the First Principle of Mathematical Induction implies the Second Principle of Mathematical Induction. (This is Exercise 4 at the end of the handout on Induction Proofs.)

   Requirements for each PMI:

   (a) PMI1 = Weak Induction
      i. The statement holds for n=1 (BASE CASE)
      ii. Whenever the statement holds for n=k, it must also hold for n=k+1 (Inductive Step)

   (b) PMI2 = Strong Induction
      i. The statement holds for n=1 (BASE CASE)
      ii. Whenever the statement holds for all $n \leq k$, it must also hold for n=k+1 (Inductive Step)

   Solution. *Proof:*
   If we assume that statement P(n) fulfills the 2 requirements for PMI2, then we can go ahead and state that statement Q(n) is "P(n) is true for all $n \leq k$*".*

   The next step is to attempt proving the statement Q(n) is true for all positive integers using PMI1 which will result in P(n) also being true for all positive integers.

   Since we assume that P(1) is true then Q(1) should also be true following the base case of PMI1. Then if the inductive step of PMI2 is also assumed to be met for P(n) then the following holds: if P(n) is true for all $n \leq k$ *then Q(n) should also be true for all* $n \leq k$ *then P(k+1) should also hold true.*

   Similarly if Q(k) is true then P(k+1) should also be true. Because P(k+1) is true then Q(k+1) should also be true since Q(k) is also true.

   So if Q(k) is true then Q(k+1) should also be true and this is the inductive step of PMI1. Since we have the Base case and Inductive step for PMI1 for Q(n) then Q(n) is true for all positive integers which also results in P(n) being true for all positive integers.

   This conclusion supports that if PMI1 holds then it is implied that PMI2 holds.